# Yousef Khan

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## Education

**University of California, Irvine** — Irvine, CA
*B.S. in Computer Science — GPA: 3.95* — *Sep 2023 – Jun 2027*

- **Graduate Coursework:** Analysis of Algorithms, Advanced Data Structures
- **Publication:** "Understanding Student Help-Seeking Behaviors in Algorithms Courses: Examining Convenience, Accessibility, and Other Themes" — *under review, SIGCSE TS 2027*

## Experience

**Software Engineering Intern** — Jun 2026 – Sep 2026
*NASA Jet Propulsion Laboratory* — *Pasadena, CA*

- Built a full-stack Grafana datasource plugin for Hermes, NASA's open-source spacecraft telemetry framework
- Shipped a TimescaleDB provider converting telemetry/EVRs to parameterized inserts on partitioned hypertables
- Added an OpenTelemetry provider exporting events to Loki and batched telemetry metrics to Prometheus
- Implemented Grafana template variable resolution with autocomplete for parameterized, reusable dashboards
- Built an Add Remote Host flow in the VS Code extension with URL/label validation

**Tech Lead, Full-Stack Developer** — Oct 2025 – Present
*Commit the Change, UCI* — *Irvine, CA*

- Leading a project team of 12 developers and 8 designers alongside 2 other leads, scoping requirements with NPOs
- Designed PostgreSQL schemas and REST APIs for roles, locations, and volunteer-location relationships
- Implemented Google/Microsoft SSO and custom auth flows with backend credential validation

**Teaching Assistant** — Mar 2024 – Apr 2026
*UCI ICS — ICS 6B/6D: Boolean Logic and Discrete Structures/Math* — *Irvine, CA*

- Verified exams, oversaw grading quality control, and made final judgments on disputes for 600+ students
- Pioneered the adoption of Pensieve, an AI grading platform, and coordinated integration into the course workflow

**Director** — May 2023 – Present
*Hack at UCI* — *Irvine, CA*

- Lead logistics for IrvineHacks, UCI's flagship hackathon (500+ attendees); direct ZotHacks (beginner hackathon)
- Designed a virtual queue check-in system using status/decision state transitions, processing 400+ day-of check-ins
- Built and debugged backend queue demotion logic across waitlist, confirmation, and waiver states

**Lead Full-Stack Developer** — Dec 2022 – Present
*Youth Spiritual Summit* — *Irvine, CA*

- Rebuilt payment system from PayPal to Stripe with embedded checkout and server-side amount validation
- Implemented idempotent checkout flow using atomic Prisma transactions over MySQL to prevent duplicate charges

## Projects

**Grouper** | *React, TypeScript, Supabase, PostgreSQL, Vercel* — Apr 2026 – Present

- Built and shipped a group coordination app (expense splitting, ride sharing, itinerary planning) to production
- Implemented optimistic UI updates backed by Supabase realtime subscriptions to keep state in sync across clients
- Built a shared expense ledger tracking $7,500+ across 100+ users, with atomic writes and server-side validation

**Smart401K** | *React, Vite, Tailwind CSS, MUI, Gemini API* — May 2025 – Present

- Collaborated with a financial expert from Intuit to build an interactive 401(k) planning tool
- Used Gemini to parse plan documents, bucket funds into asset categories, and select holdings by expected return
- Generated dynamic investor profiles driving allocation recommendations, visualized with MUI charts

**ICSYAK** | *Python, discord.py, Flask, Oracle Cloud, Slurm* — Oct 2023 – Present

- Developed a Discord bot integrating online class forums (EdStem, Piazza) for 1000+ ICS students across 8 courses
- Implemented time-based and on-demand thread lookup by ID, class, and other keys via edapi/piazza-api
- Contributed features to a Python package to pull template PDFs from Gradescope into Discord

## Technical Skills

**Languages**: Python, C, C++, Java, JavaScript/TypeScript, Go, SQL
**Frameworks**: React, Next.js, Vite, Tailwind CSS, Flask, FastAPI, Prisma
**Technologies**: Git, GitHub Actions, Docker, GCP, PostgreSQL, MySQL, Supabase, Stripe, Grafana, OpenTelemetry